

Exploring the diameter of a hypothetical rubber ball at photon scale, relativistic effects, and impact deformation.



Photon-Sized Rubber Ball

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What is a "Photon-Sized" Ball?

Photons are elementary particles that are considered point-like with no measurable size in the classical sense. However, we can associate a size with a photon through its wavelength (λ), which depends on its energy.

For example, a photon of visible light (say, green light at 550 nm wavelength) has a wavelength of:

$$\lambda = 550 \times 10^{-9} \text{ m} = 550 \text{ nanometers}$$

If we imagine a rubber ball with a diameter equal to this wavelength, its diameter would be:

$$D_{\text{e}} = 550 \text{ nm} = 0.55 \text{ micrometers}$$

Note: A rubber ball this small is not physically feasible to manufacture, but we can still explore the physics.

Other Lights: Across the Electromagnetic Spectrum

If we consider different types of light (different wavelengths), the "size" of a photon-sized rubber ball would vary dramatically across the electromagnetic spectrum:

Type of Light	Typical Wavelength	Diameter of Ball
Radio Waves	1 m – 100 km	1 m – 100 km (huge!)
Microwaves	1 mm – 1 m	1 mm – 1 m
Infrared	700 nm – 1 mm	0.7 – 1000 μm
Visible Light	400 – 700 nm	0.4 – 0.7 μm
Ultraviolet	10 – 400 nm	0.01 – 0.4 μm
X-Rays	0.01 – 10 nm	0.01 – 10 nm

Type of Light	Typical Wavelength	Diameter of Ball
Gamma Rays	< 0.01 nm	< 0.01 nm

Note: These diameters range from incredibly large (radio waves) to incredibly small (gamma rays). A rubber ball of such extreme sizes is not feasible, but the exercise illustrates the vast scale of the electromagnetic spectrum.

Relativistic Length Contraction

When an object moves at relativistic speeds (close to the speed of light), it undergoes **length contraction** in the direction of motion as observed by a stationary observer. The formula is:

$$L = L_0 \times \sqrt{1 - v^2/c^2}$$

Where:

- L = contracted length (diameter in direction of motion)
- L_0 = rest diameter (550 nm in our example)
- v = velocity of the object
- c = speed of light ($\approx 3 \times 10^8$ m/s)

Example: If our rubber ball moves at 90% the speed of light ($v = 0.9c$):

$$L = 550 \text{ nm} \times \sqrt{(1 - 0.9^2)} = 550 \text{ nm} \times \sqrt{0.19} \approx 550 \text{ nm} \times 0.436 \approx 240 \text{ nm}$$

The diameter appears contracted to about 240 nm in the direction of motion. Note: The ball becomes an oblate spheroid (squashed sphere) to the observer.



Deformation from Impact

When a rubber ball impacts a surface at high speed, it deforms due to the forces involved. The deformation depends on:

- Impact speed
- Material properties (Young's modulus, Poisson's ratio)
- Ball diameter and wall thickness (if hollow)
- Surface properties

For a spherical elastic ball impacting a rigid surface, the maximum deformation (δ) can be approximated by Hertzian contact theory :

$$\delta = \left(\frac{3F(1-\nu^2)}{4E\sqrt{R}} \right)^{2/3}$$

Where:

- F = impact force
- ν = Poisson's ratio of the material (≈ 0.5 for rubber)
- E = Young's modulus of the material ($\approx 0.01\text{--}0.1$ GPa for rubber)
- R = radius of the ball

Note: At photon scale, quantum effects and surface forces would dominate, making classical continuum mechanics inapplicable. This formula is for macroscopic rubber balls.

Higher impact speeds generally lead to greater deformation, up to the point where the ball may break or undergo plastic deformation.



Putting It Together

For our hypothetical photon-sized rubber ball:

1. **Rest diameter:** ~ 550 nm (based on green photon wavelength)
2. **Relativistic effect:** At high speeds, the diameter contracts in the direction of motion.
3. **Impact deformation:** At macroscopic scales, higher speed causes more deformation—but at nanoscale, different physics applies.

This thought experiment highlights the importance of scale in physics: quantum effects dominate at nanoscale, relativistic effects at high speeds, and continuum mechanics for everyday objects.

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